



Semantic-Based Implicit Feature Transform for Few-Shot Classification

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Abstract

Few-shot learning aims to recognize instances from previously unseen classes based on a very limited number of examples. However, models often face the challenge of overfitting due to the biased distribution computed from extremely scarce training data. This work proposes a Semantic-based Implicit Feature Transform (SIFT) method to implicitly generate high-quality features for few-shot learning tasks. In this method, we employ an encode-transform-decode pipeline to facilitate the direct transfer of feature instances from base classes to novel classes via semantic transformation, ensuring the generation of semantically meaningful features. Additionally, a compactness constraint is imposed on the generated features, with novel class prototypes serving as anchors, to ensure that the features are distributed around the class centers. Furthermore, we integrate the cluster centers of the query set with the initial prototypes computed from the support set to produce less biased class prototypes, which can serve as better anchors for feature reconstruction. The experimental results reveal that our method outperforms the baselines by substantial margins and achieves state-of-the-art few-shot classification performance on the miniImageNet and CIFAR-FS datasets in both inductive and transductive settings, demonstrating the superiority of our method. Furthermore, the comprehensive ablation studies provide additional validation of its effectiveness. The code is available at: <https://github.com/pmhDL/SIFT.git>.

Keywords Implicit feature transform · Tied weight · Cluster assignments · Path planning problem

1 Introduction

Deep learning has demonstrated remarkable performance across various computer vision tasks, primarily relying on extensive supervision. However, in certain real-world applications, acquiring a substantial amount of well-annotated training data is impractical (Vartak et al., 2017; Altae-Tran

et al., 2017). This challenge motivates researchers to explore the data-scarce technique known as few-shot learning (FSL), designed to learn new concepts from very few labeled examples. FSL methods leverage experiences from similar tasks based on the episodic paradigm, equipping artificial intelligence with a human-like ability to continuously acquire novel concepts (He et al., 2015).

Data augmentation is a straightforward technique to address the challenge of extremely limited data. Naturally, as the number of labeled samples grows, the data distribution can gradually approach the ground truth distribution. However, most existing methods perform data augmentation in the visual feature space, using the few labeled support samples as anchors, which typically results in marginal performance improvements. In (Yang et al. 2020), a Distribution Calibration (DC) method is introduced to adjust the distributions of the novel classes by transferring distribution knowledge from base classes, where the few labeled novel samples serve as anchors. However, even with this calibration, the distribution remains constrained by the limited number of labeled samples. As illustrated in Fig. 1a, the augmented samples in the

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visual space distribute around the support samples and can not cover the ground truth distribution adequately.

To address this challenge, we propose the Semantic-based Implicit Feature Transform (SIFT) method to implicitly generate high-quality features for few-shot learning tasks. Our approach draws inspiration from the concept proposed in DC method, which leverages knowledge from similar base classes to augment features for novel classes. Additionally, our approach is motivated by the observation in Fig. 1b that samples from distinct but semantically similar classes often exhibit visually similar representations. A key advantage of our method lies in its reliance on class-specific semantic embedding, which remains unique and fixed for each class regardless of the number of labeled samples. This inherent stability ensures that the augmented features transferred from the base classes are less biased. Our method differs from the DC method in several aspects: (1) Our method selects base classes based on semantic similarities, which involves optimizing a decision variable based on a 0–1 linear programming model to prevent novel classes from selecting duplicate base classes. Whereas, the DC method selects base classes based on visual similarities between the average visual features of the base classes and the support features of the novel task, without constraints on the selected classes. (2) Our method explicitly transfers feature instances through semantic transformation, with class-specific semantic embeddings serving as anchors. This process is agnostic to the number of labeled samples. Conversely, the DC method transfers the distributions of the base classes to the novel classes based on the few labeled support features, which is dependent on the “shot” of the task. (3) Our method generates new features by directly transferring feature instances from base classes, while DC method samples new features from the calibrated distributions. Throughout this paper, the term “semantic embedding” consistently refers to the word embedding of the class label, which is computed using the GloVe (Pennington et al., 2014) method.

Our approach employs an encode-transform-decode pipeline to facilitate the transfer of features from base classes to novel classes, capitalizing on the intrinsic semantic relationships between these categories. This process imposes constraints on the transformation between corresponding classes, with semantic embeddings serving as essential anchors. These constraints ensure the generation of semantically meaningful latent features, contributing to the reconstruction of more realistic visual features. Previous works have explored the concept of using encoder and decoder models to augment samples. For instance, DeVries and Taylor (2017) apply simple transformations, such as adding noise or interpolating features, in a latent embedding space, and then reconstruct the augmented features to the original embedding space. Schwartz et al. (2018) propose a Delta-encoder method that extracts transferable intra-class deltas between same-class pairs of training examples and applies these deltas to the few samples of the novel classes. Both methods in (DeVries and Taylor 2017) and (Schwartz et al. 2018) employ the encoder-decoder paradigm to augment samples by projecting features into a low-dimensional latent embedding space and applying perturbations to generate diverse features. However, the diversity of features can be limited, as they use the few labeled novel samples as anchors.

To reduce the bias of the anchor, we propose a novel prototype rectification method based on a shortest path optimization strategy. Previous works have explored some prototype rectification methods. For instance, MetaNODE (Zhang et al., 2022) formulates prototype rectification as a prototype optimization problem, using ordinary differential equations (ODEs) for meta-optimization on the mean prototypes. This approach involves a complex meta-learning process. Liu et al. (2020) propose a BD-CSPN method, which rectifies prototypes by integrating pseudo-labeled query features with the support features. However, the pseudo-labels are determined based on the similarity to the initial prototypes, which may introduce bias into the pseudo-labels, thereby negatively impacting prototype rectification. In con-

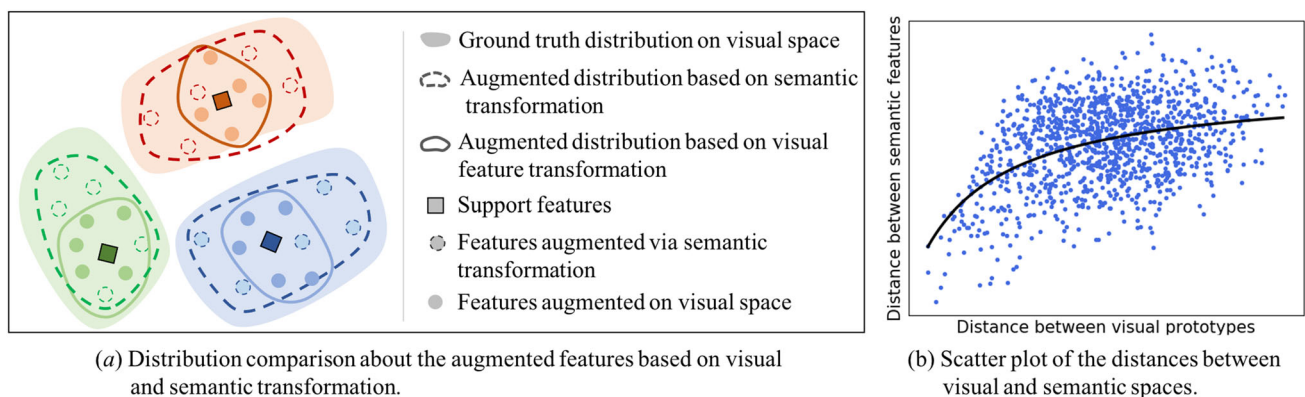


Fig. 1 Conceptual description of SIFT

trast to these methods, our approach rectifies prototypes by incorporating the cluster centers of the query samples. The clustering process relies on the data distribution of the query samples, which is agnostic with the few labeled support samples, ensuring that the cluster centers do not introduce biased information from support samples. Subsequently, we assign labels to the cluster centers based on a global optimization strategy, reducing the risk of incorrect assignments.

To ensure the discriminability of the augmented samples, we impose a compactness constraint (Zhu et al., 2019) on the generated features to encourage high intra-class compactness for each novel class. Compactness is defined as the distance between the prototypes and the generated features. By using the prototypes as anchors, the generated features distribute around the prototypes of their respective classes. The model is trained in an end-to-end fashion. Finally, we pass the selected base features through the trained networks, which include the encoder, transformation network, and decoder, in sequence to generate discriminative features for the novel classes. The main contributions are summarized as follows:

- 1) We introduce a semantic-based implicit feature transformation approach that explicitly transfers features from base classes to novel tasks.
- 2) We propose a shortest path optimization strategy to refine the prototypes, which integrates the cluster centers of the query features with the initial prototypes derived from the support set to reduce prototype bias.
- 3) Comprehensive ablation studies and extensive experiments validate the effectiveness of our data augmentation strategy and its capacity for generalization to other few-shot learning baselines.

The remainder of this paper is organized as follows: Sect. 2 provides a concise overview of related works. In Sect. 3, we initially introduce the problem definition and then present our Semantic-based Implicit Feature Transform (SIFT) method. Section 4 is dedicated to extensive experiments that confirm the effectiveness of our SIFT method, including insightful ablation studies and in-depth analyses. Lastly, we draw our conclusions in Sect. 5.

2 Related Work

2.1 Few-Shot Learning Methods

Meta learning is an effective few-shot learning paradigm that leverages experiences from similar tasks based on the episodic formulation. Previous researches on few-shot learning are made towards the following aspects. (1) *Metric learning methods* aim to learn a good metric space, where test samples can be classified via a simple metric rule. Works

include nearest neighbor classifier with Euclidean (Snell et al., 2017) or cosine distance (Vinyals et al., 2016), learnable relation network-based method (Sung et al., 2018) and task specific metric-based method for respective task (Oreshkin et al., 2018). (2) *Gradient-based methods* aim to learn models that can generalize well to unseen tasks with limited supervisions. Methods, such as MTL (Sun et al., 2019), MAML (Finn et al., 2017) and LEO (Rusu et al., 2018) fall into this category. (3) *Model-based methods* rely on the properties of specific model architectures, such as recurrent and memory-augmented networks (Mishra et al., 2018; Munkhdalai & Yu, 2017; Santoro et al., 2016).

2.2 Few-Shot Learning Based on Data Augmentation

Data augmentation-based few-shot learning methods try to synthesize data or features by learning a generative model to alleviate the data insufficiency problem. Previous methods usually apply content preserving transformation techniques on the input images, which fail to ensure the diversity of the generated samples. Recently, instead of synthesizing new image instances, a number of effective augmentation methods have been proposed in visual or semantic feature levels. Hallucination-based methods (Wang et al., 2018, 2019; Hariharan & Girshick, 2017; Schwartz et al., 2018; Chen et al., 2019; Yang et al., 2020) augment the training data by hallucinating additional examples for the unseen classes. Hariharan and Girshick (2017) present a way of “hallucinating” additional examples for novel classes by transferring modes of variation from the base classes. Schwartz et al. (2018) propose an auto-encoder to augment samples by encoding the intra-class deformations. Chen et al. (2019) propose to directly synthesize instance features by leveraging semantics using a novel auto-encoder network. Yang et al. (2020) introduce a distribution calibration strategy to generate features for the novel tasks, which does not need extra learnable parameters. However, this method requires lots of augmented samples to achieve the improved performance, while our method does not demand many augmented samples, which greatly reduces the computational burden and computational time. Our SIFT method augments samples based on semantic transformation, which won’t be limited by the few labeled samples. A simple classifier trained with the features generated by our SIFT method along with the support data can achieve considerable performance improvement.

2.3 Few-Shot Learning Based on Multimodal Features

Due to the limited availability of samples in few-shot learning tasks, researchers have resorted to utilizing multimodal information to maximize knowledge utilization in such scenarios. Among these modalities, semantic embeddings have

been widely employed. Xing et al. (2019) introduce an Adaptive Modality Mixture Mechanism (AM3), which enhances prototype representations by combining visual prototypes with class semantic features. Li et al. (2020) leverage semantic similarities between classes as adaptive margins for improved sample differentiation. Schwartz et al. (2022) propose a method employing multiple semantics to generate more representative prototypes, mirroring the way infants learn. Chen et al. (2019) propose an approach to augment visual features in the semantic space by generating samples from class semantic Gaussian distributions. Xu et al. (2022) propose a Variational AutoEncoder (VAE)-based feature generation approach conditioned on class semantic embeddings to construct class-representative prototypes. In a recent development by Pan et al. (2023), a task-adaptive semantic feature learning approach is introduced to complement absent modalities in query samples. All these methods primarily focus on enhancing representations for either classes or individual sample points. In contrast, our approach diverges from this paradigm by aiming to augment features through the transfer of base feature instances to novel classes via semantic transformation. Our method generates new features based on feature instances rather than class distributions, making the generated features more closely aligned with real features.

3 Method

3.1 Few-Shot Setting

In few-shot learning, the standard N -way K -shot episodic training paradigm is adopted to learn the model. We define the training dataset as $\mathcal{D}_{base} = \{(x_i^b, y_i^b) | y_i^b \in \mathcal{C}_{base}\}$, which contains abundant labeled samples. Also, we define the test dataset as $\mathcal{D}_{novel} = \{(x_i^n, y_i^n) | y_i^n \in \mathcal{C}_{novel}\}$, which is used during the test stage. Categories of $\mathcal{D}_{novel}$ and $\mathcal{D}_{base}$ are disjoint from each other, i.e. $\mathcal{C}_{base} \cap \mathcal{C}_{novel} = \emptyset$. The model is trained and tested in task-wise manner, where each task consists a support set $\mathcal{S} = \{(x_i^n, y_i^n, s_{y_i^n}^n)\}_{i=1}^{N \times K}$ and a query set $\mathcal{Q} = \{(x_i^n)\}_{i=1}^{N \times L}$. $x_i \in \mathbb{R}^d$ is the feature vector of an image and y_i is the class label of x_i . $s_{y_i^n}^n$ is the semantic embedding of class y_i^n , which is the word embedding of the class label computed with GloVe method. The goal of our method is to augment the support set by transferring samples from base dataset to the novel tasks. Then a classifier can be trained with the augmented samples and the support samples. Finally, the classification performance is evaluated over lots of testing tasks.

3.2 Methodology

The framework of our SIFT approach is illustrated in Fig. 2. The key idea is to transfer base data to novel tasks through semantic space in a task adaptive manner to augment support set. This process includes three steps. Firstly, an encoder projects the visual features of base classes to semantic space. Then the transformation network transfers semantic embeddings from base classes to novel classes. Afterwards, a decoder projects the semantic embeddings back to visual space. To preserve all the information contained in the original visual features and generate semantically meaningful visual features, we impose a reconstruction constraint, that is modeled as the ridge regression loss (Kodirov et al., 2017) to the encoder and decoder. The ground truth semantic embeddings of the novel classes work as the anchor for the visual to semantic projection. These three modules are trained in an end-to-end manner. With the trained networks, we feed samples of base classes through the encoder, transformation network and decoder in turn to generate extra samples for the novel classes.

3.2.1 Base Classes Selection Based on 0–1 Linear Programming

We select related base classes for the novel task based on similarities between the semantic embeddings of the base and novel classes. Note that the semantic embeddings in base dataset are denoted as $S^b = \{s_j^b | j = 1, \dots, |\mathcal{C}_{base}|\}$, and that in a novel task are denoted as $S^n = \{s_i^n | i = 1, \dots, N\}$. We first measure the cosine similarity between the base and novel semantic embeddings as $p_{ij} = \frac{s_i^n s_j^b}{\|s_i^n\| \cdot \|s_j^b\|}$, where $i = 1, \dots, N, j = 1, \dots, |\mathcal{C}_{base}|$. We select base class with the greatest similarity to the novel class. To avoid the situation that two or more novel classes sharing the same base class due to the similar semantic embeddings of different novel classes, we construct the class selection process as a 0–1 linear programming model.

$$\begin{aligned}
 & \text{maximize} \quad \sum_{i=1}^N \sum_{j=1}^{|\mathcal{C}_{base}|} p_{ij} z_{ij} \\
 & \text{subject to} \quad \sum_{j=1}^{|\mathcal{C}_{base}|} z_{ij} = 1 \quad \text{for } i = 1, \dots, N \\
 & \quad \sum_{i=1}^N z_{ij} \leq 1 \quad \text{for } j = 1, \dots, |\mathcal{C}_{base}| \\
 & \quad z_{ij} \in \{0, 1\}
 \end{aligned} \tag{1}$$

where $i = 1, \dots, N, j = 1, \dots, |\mathcal{C}_{base}|$. z_{ij} is a 0-1 decision variable, that controls which base class to be selected. In

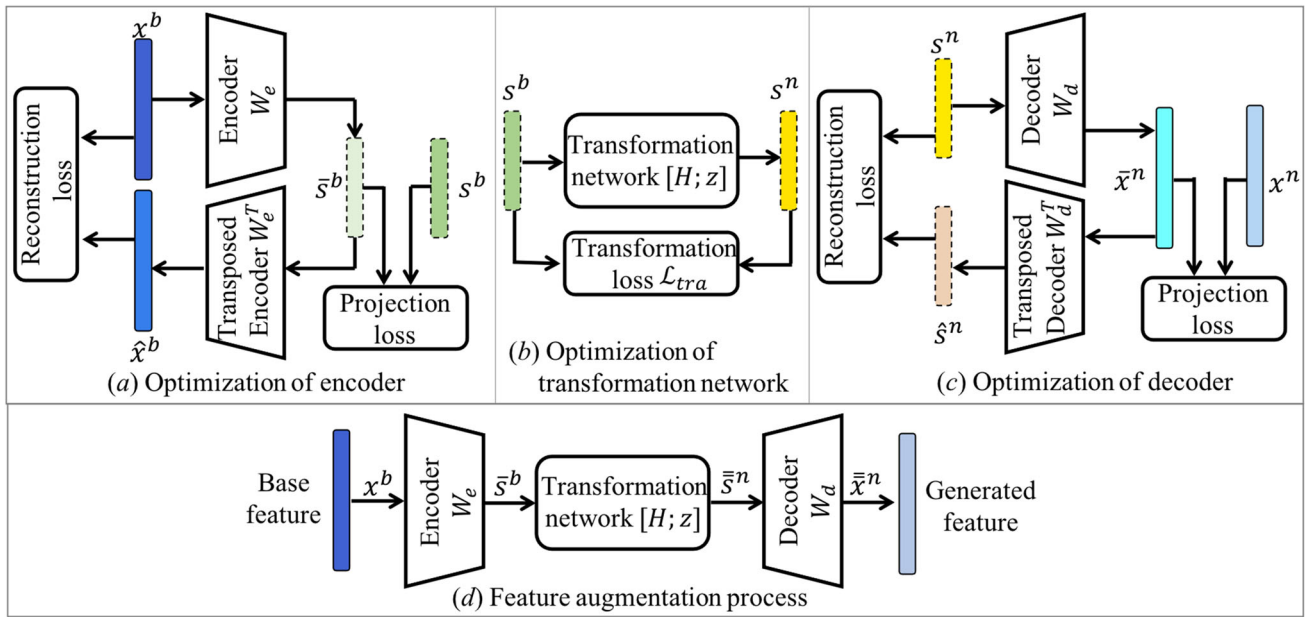


Fig. 2 The framework of our SIFT method. x^b and s^b are the visual features and semantic embeddings from the selected base classes, while x^n and s^n are the ones from the novel classes. $\hat{x}^b$ and $\bar{s}^b$ denote the reconstructed visual feature and the projected semantic embedding, while $\bar{x}^n$

and $\hat{s}^n$ are the projected visual feature and the reconstructed semantic embedding. $\bar{s}^n$ is the transformed semantic embedding and $\bar{x}^n$ is the generated novel visual feature

model Eq. (1), the objective function maximizes the sum of similarities between the novel classes and the selected base classes, followed with two constraints on the decision variable. a) the sum of each row in z must be 1. b) the sum of each column in z must be less than or equal to 1. The real meaning of constraint a) denotes that each novel class must and only be assigned with one base class, and the constraint b) denotes each base class must be only selected at most once. By solving this model, we can get the optimal decision variable that matches non-duplicate base classes for the novel classes. The selected classes are denoted as $I = \{i_{z_i=1} | i = 1, \dots, N\}$, $\mathbb{I}_{condition}$ is a indicative function which returns the index of the vector when the condition is met. Then we choose K' samples per selected base class.

$$X^{bs} = \bigcup_{j \in I} \{(x^b, y^n) | x^b \in \mathbb{S}(\mathcal{D}_{base}^j, K'), y^n = index(I, j)\} \quad (2)$$

where $\mathbb{S}(\cdot, \cdot)$ is a sample selection operator, $index(I, j)$ is an operation to return the order of j in I , i.e. the corresponding label of the base class j in the novel task. $|X^{bs}| = N \times K'$. To encourage diversity of the generated samples, we select samples in a random way. We denote the semantic embeddings of the selected base classes as $S^{bn} = \{s_i^b | i \in I\}$.

3.2.2 Visual-Semantic Encoder Formulation

We formulate the visual-to-semantic projection function as a regression loss, and denote x_i^b as the visual feature of i^{th} sample from X^{bs} and s_i^b as the ground truth semantic embedding of x_i^b . To ensure the encoder project the visual features to their respective corresponding semantic embeddings and preserve all the information contained in the original visual features, we impose a projection constraint and a reconstruction constraint to the encoder. The constraints are formulated as $\mathcal{L}_{enc}$.

$$\min_{W_e} \mathcal{L}_{enc} = \sum_{i=1}^{N \times K'} \|s_i^b - \bar{s}_i^b\|_2^2 + \|x_i^b - \hat{x}_i^b\|_2^2 \quad (3)$$

$$s.t. \quad \bar{s}_i^b = W_e x_i^b \quad \hat{x}_i^b = W_e^T \bar{s}_i^b$$

where $\bar{s}_i^b$ is the projected semantic embedding and $\hat{x}_i^b$ is the reconstructed visual feature, W_e denotes the trainable parameter of the encoder and W_e^T is the transposed one. By training with weight tying (Ranzato et al., 2007), the desire is to learn a set of weights where the property $W_e^T = W_e^\dagger$ is true.

3.2.3 Feature Transformation from Base to Novel Classes

In this section, we aim to learn a transformation network $g(\cdot; [H, z]): s^b \rightarrow s^n$, which is optimized by minimizing the Euclidean distance between the novel semantic embeddings

s^n and the generated semantic embeddings $\bar{s}^n$. We denote this objective as the transformation loss.

$$\min_{[H,z]} ||s^n - \bar{s}^n||_2^2 \quad s.t. \quad \bar{s}^n = Hs^b + z \quad (4)$$

$$\mathcal{L}_{tra} = ||s^n - \bar{s}^n||_2^2$$

3.2.4 Semantic-Visual Decoder Formulation

The decoder employs similar optimization process as the encoder. The difference is that the decoder reverses the projection direction: from semantic space to visual space. Given a support set $\mathcal{S} = (x_i^n, y_i^n)_{i=1}^{N \times K}$ and the corresponding semantic embedding set $S^n = \{s_i^n | i = 1, \dots, N\}$, we project the semantic embeddings to visual space with the support features as the reference points. The decoder is constrained by minimizing the distance between the projected visual features $\bar{x}^n$ and the novel visual features x^n from support set. To ensure the decoder generate semantically meaningful visual features, we further impose a reconstruction constraint to the decoder by minimizing the distance between the ground truth semantic embeddings s^n of the novel classes and the reconstructed semantic embeddings $\hat{s}^n$. These constraints are formulated as $\mathcal{L}_{dec}$.

$$\min_{W_d} \mathcal{L}_{dec} = \sum_{i=1}^{N \times K} ||x_i^n - \bar{x}_i^n||_2^2 + ||s_i^n - \hat{s}_i^n||_2^2 \quad (5)$$

$$s.t. \quad \bar{x}_i^n = W_d s_i^n, \hat{s}_i^n = W_d^T x_i^n$$

where W_d is the parameter of the decoder and W_d^T denotes the transposed one. We consider tied weights (Kodirov et al., 2017) for the decoder, which has the property of $W_d^T = W_d^\dagger$.

3.2.5 Compactness Constraint to the Generated Features

Given a novel task $\mathcal{T} = \{\mathcal{S}, \mathcal{Q}\}$, where $\mathcal{S} = (x_i^n, s_i^n, y_i^n)_{i=1}^{N \times K}$, we select N base classes for the novel task by the way in Sect. 3.2.1. The selected dataset is denoted as $X^{bs} = \{(x_i^b, y_i^n)\}_{i=1}^{N \times K'}$. We generate novel visual features $\bar{x}_i^n$ from X^{bs} and augment the support set as follow:

$$\bar{x}_i^n = W_d(HW_e x_i^b + z), \quad \mathcal{S}_{aug} = \{(\bar{x}_i^n, y_i^n)\}_{i=1}^{N \times K'} \quad (6)$$

where K' is the number of generated features for each novel class. In order to get discriminative features, we impose a compactness constraint to the generated features, which is formulated as follow:

$$\mathcal{L}_{cpt} = \frac{1}{|X^{bs}|} \sum_{j=1}^N \sum_{y_i=j} ||\bar{x}_i^n - m_j||_2^2 \quad (7)$$

where m_j denotes the prototype of class j , which works as the reference point for the generated features. This constraint encourages the generated features of the same class gathering around the class centers, which improves the discrimination of the generated features.

3.3 Prototype Rectification

Our method can be implemented in inductive and transductive settings. For the inductive setting, we compute the prototypes by averaging the labeled features of each class. For the transductive setting, we propose a novel prototype rectification method by incorporating the cluster centers of the query samples with the initial prototypes computed from support set. Firstly, the query samples are clustered into N clusters with a simple clustering algorithm K -means. Since clustering is an unsupervised classification method, the clusters themselves cannot tell which category they belong to. To infer class label for each cluster, we model the label assignment problem as the path planning problem, which is formulated as Eq. (8).

$$\begin{aligned} & \text{minimize} \quad \sum_{i=1}^N \sum_{j=1}^N d_{ij} w_{ij} \\ & \text{subject to} \quad \begin{cases} \sum_{i=1}^N w_{ij} = 1, \quad j = 1, \dots, N \\ \sum_{j=1}^N w_{ij} = 1, \quad i = 1, \dots, N \\ w_{ij} \in \{0, 1\} \end{cases} \end{aligned} \quad (8)$$

where $d_{ij} = ||m_i - c_j||_2^2$, $i = 1, \dots, N$, $j = 1, \dots, N$ is a distance matrix between the initial prototypes and the cluster centers. w_{ij} is a binary decision variable that decides which class the cluster belongs to. m_i denotes the i^{th} initial prototype and c_j denotes the j^{th} cluster center. The objective is to minimize the total distance between the matched prototypes and cluster centers. The constraints ensure that each class label must be only assigned to one cluster center. This model can produce global optimal solution for the label assignment, thus avoid the situation that two or more clusters share a common class label due to the close distribution of samples from different classes. Then we compute the rectified prototype of k^{th} class as follow:

$$\hat{m}_k = \frac{m_k + c_{\mathbb{I}(w[k,:]=1)}}{2} \quad (9)$$

where $\mathbb{I}(\cdot)$ serves as an indicator function that returns the position of the nonzero value. This prototype rectification method can effectively mitigate bias in the initial prototypes.

3.4 Train the SIFT Model and the Classifier

Overall, the proposed model is trained in an end-to-end manner with the joint loss as follow:

$$\mathcal{L} = \mathcal{L}_{cpt} + \lambda_1 \mathcal{L}_{enc} + \lambda_2 \mathcal{L}_{tra} + \lambda_3 \mathcal{L}_{dec} \quad (10)$$

where the balance factors λ_1 , λ_2 and λ_3 are consistently set to 1 in all experiments. Now that the feature generation model has been trained, it is used to generate features to augment the support set, see Eq. (6). Then we train a task-specific classifier based on the generated features and the original support set. The training task follows the standard cross-entropy loss optimization:

$$\min_{\theta} \sum_{(x,y) \in \mathcal{S} \cup \mathcal{S}_{aug}} -\log p(y|x; \theta) \quad (11)$$

where θ is the parameter of the classifier. The trained classifier is then used to predict labels for the query samples. We summarize the process of our method in Algorithm 1.

Algorithm 1 Proposed SIFT algorithm

Require: Base set $\mathcal{D}_{base}$, novel task $\mathcal{T} = \{\mathcal{S}; \mathcal{Q}\}$, iteration T .

Ensure: The generated novel features.

- 1: Select N base classes for the novel task;
 - 2: Sample K' features from each selected base class by Eq. (2);
 - 3: **for** t in 1 to T **do**
 - 4: Impose projection and reconstruction constraints to the encoder by Eq. (3)
 - 5: Optimize the feature transformation network $g(\cdot; [H, z])$ by Eq. (4)
 - 6: Impose projection and reconstruction constraints to the decoder by Eq. (5)
 - 7: Impose compactness constraint to the generated features by Eq. (7)
 - 8: Optimize parameters $\{W_e; [H, z]; W_d\}$ of the model by minimizing Eq. (10)
 - 9: **end for**
-

4 Experiments

4.1 Dataset

We evaluate our SIFT method on miniImageNet (Ravi & Larochelle, 2016), tieredImageNet (Ren et al., 2018), Caltech-UCSD Birds 200 (Welinder et al., 2010) (CUB) and CIFAR-FS (Chen et al., 2018) benchmarks.

miniImageNet is a subset of ILSVRC-2012 (Russakovsky et al., 2015), which consists of 100 classes that are split into 64, 16 and 20 classes for training, validation and test respectively. Each class contains 600 images of size $84 \times 84 \times 3$.

tieredImageNet is a larger subset of ILSVRC-12 dataset (Russakovsky et al., 2015), which contains 608 classes sampled from hierarchical category structure. The average number of images in each class is 1281 of size $84 \times 84 \times 3$. We use 351, 97, and 160 classes for training, validation, and test, respectively.

CIFAR-FS is a dataset derived from CIFAR-100 with images of size $32 \times 32 \times 3$. It contains 100 categories with 600 instances in each class. All the classes are split into 64, 16 and 20 for training, validation and test.

Caltech-UCSD Birds 200 is a fine-grained dataset consisting of 200 classes of birds with a total number of 11,788 images of size $84 \times 84 \times 3$. Following the split in previous work (Chen et al., 2018), we use 100, 50 and 50 classes for training, validation and test, respectively. This dataset also provides 312-dimensional semantic attribute vectors on a per-class level.

4.2 Network Architecture

We use two popular networks: the ResNet-12 and WideResNet (WRN-28-10) (Zagoruyko & Komodakis, 2016) to extract features for all the datasets. We first pre-train the feature extractor on the base dataset. The trained extractor outputs 640-dimensional visual feature vectors for the images. The encoder is constructed with a fully connected layer without the bias term, the transformation network is modeled as a fully connected layer, and the decoder is implemented as a fully connected layer without the bias term as well.

4.3 Implementation Details

At the model training stage, Adam is used as the optimizer. The initial learning rate is 0.001 and decreased to 0.2 times of the previous stage per 10 steps. The model is trained for 100 iterations per task. During the test stage, we use the logistic regression (LR) and the support vector machine (SVM) as the classifier. The number of generated features for different datasets are determined by the experiment in Sect. 4.4.2. The visual features are extracted by the pre-trained network, and the semantic embeddings are computed by the Glove method in an average manner of the word embeddings of each class.

4.4 Results and Analysis

4.4.1 Comparison to the State-of-the-art Methods

To evaluate the performance of our SIFT method, we conduct a comprehensive comparison with state-of-the-art methods. Tables 1 and 2 display the average classification accuracies across 600 novel tasks in both 5-way 1-shot and 5-way 5-shot scenarios, with each task comprising 15 randomly

Table 1 The few-shot classification accuracy results on miniImageNet and tieredImageNet datasets

Model	Backbone	miniImageNet 5-way		tieredImageNet 5-way	
		1-shot	5-shot	1-shot	5-shot
In.					
MatchingNet Vinyals et al. (2016)	ResNet-12	63.08 ± 0.80	75.99 ± 0.60	68.50 ± 0.92	80.60 ± 0.71
ProtoNet Snell et al. (2017)	ResNet-12	60.37 ± 0.83	78.02 ± 0.57	65.65 ± 0.92	83.40 ± 0.65
Curvature Generation Gao et al. (2021)	ResNet-12	67.02 ± 0.20	82.32 ± 0.14	71.66 ± 0.23	85.50 ± 0.15
Meta Navigator Zhang et al. (2021)	ResNet-12	67.14 ± 0.80	83.82 ± 0.51	74.58 ± 0.88	86.73 ± 0.61
R-SVAE* Xu and Le (2022)	ResNet-12	74.84 ± 0.23	83.28 ± 0.40	76.98 ± 0.65	85.77 ± 0.50
AM3-TADAM* Xing et al. (2019)	ResNet-12	65.30 ± 0.49	78.10 ± 0.36	69.08 ± 0.47	82.58 ± 0.31
TRAML* Li et al. (2020)	ResNet-12	67.10 ± 0.52	79.54 ± 0.60	—	—
TriNet* Chen et al. (2019)	ResNet-18	58.12 ± 1.37	76.92 ± 0.69	—	—
MultiSem* Schwartz et al. (2022)	Dense-121	67.30	82.10	—	—
FSLKT* Peng et al. (2019)	ConvNet128	64.42 ± 0.72	74.16 ± 0.56	—	—
Centroid alignment Afrasiyabi et al. (2020)	WRN-28-10	65.92 ± 0.60	82.85 ± 0.55	74.40 ± 0.68	86.61 ± 0.59
MetaQDA Zhang et al. (2021)	WRN-28-10	67.83 ± 0.64	84.28 ± 0.69	74.33 ± 0.65	89.56 ± 0.79
TasNet* Pan et al. (2023)	WRN-28-10	75.34 ± 0.71	86.11 ± 0.51	72.04 ± 0.85	86.70 ± 0.58
DC+LR Yang et al. (2020)	WRN-28-10	68.57 ± 0.55	82.88 ± 0.42	78.19 ± 0.25	89.90 ± 0.41
SIFT+LR* (ours)	WRN-28-10	77.31 ± 0.67	86.95 ± 0.53	77.86 ± 0.77	89.65 ± 0.52
SIFT+SVM* (ours)	WRN-28-10	76.66 ± 0.69	86.19 ± 0.47	77.47 ± 0.81	89.16 ± 0.55
Tran.					
ICI Wang et al. (2020)	ResNet-12	72.39 ± 0.62	83.27 ± 0.33	77.48 ± 0.62	86.84 ± 0.36
POODLE Le et al. (2021)	ResNet-12	77.56	85.81	79.67	86.96
ProtoComNet* Zhang et al. (2021)	ResNet-12	73.13 ± 0.85	82.06 ± 0.54	81.04 ± 0.89	87.42 ± 0.57
SSR Shen et al. (2021)	WRN-28-10	72.40 ± 0.60	80.20 ± 0.40	79.50 ± 0.60	84.80 ± 0.40
AIM Lee et al. (2021)	WRN-28-10	71.22 ± 0.57	82.25 ± 0.34	—	—
LaplacianShot Ziko et al. (2020)	WRN-28-10	74.86 ± 0.19	84.13 ± 0.14	80.18 ± 0.21	87.56 ± 0.15
TIM Boudiaf et al. (2020)	WRN-28-10	77.80	87.40	82.10	89.80
TasNet* Pan et al. (2023)	WRN-28-10	78.62 ± 0.67	87.75 ± 0.45	77.94 ± 0.97	88.27 ± 0.60
BD-CSPN Liu et al. (2020)	WRN-28-10	70.31 ± 0.93	81.89 ± 0.60	78.74 ± 0.95	86.92 ± 0.63
SIFT+LR* (ours)	WRN-28-10	81.03 ± 0.71	87.80 ± 0.47	82.66 ± 0.82	90.03 ± 0.54
SIFT+SVM* (ours)	WRN-28-10	80.47 ± 0.75	87.36 ± 0.49	81.93 ± 0.89	89.96 ± 0.53

The best results are highlighted in bold

They present the mean accuracy on 600 novel episodes with a 95% confidence interval

In. and **Tran.** denote the inductive and transductive settings, respectively

“—” signifies the result is unavailable. Semantic-based methods are marked with *

sampled query samples per class. We employ both logistic regression (LR) and support vector machine (SVM) as classifiers to demonstrate the generalizability of our SIFT method across different classifiers. From Tables 1 and 2, we can observe that our SIFT method consistently outperforms the state-of-the-art few-shot classification methods in miniImageNet and CIFAR-FS datasets, in both inductive and transductive settings. Furthermore, in the tieredImageNet and CUB datasets, our method surpasses the state-of-the-art methods in the transductive setting and the inductive setting, respectively. Notably, the performance gain is particularly significant in the 1-shot case, highlighting the reliability of our data augmentation method, which excels in handling extremely low-shot classification problems.

4.4.2 Impact of the Number of Generated Features

Figure 3 shows the analysis about the classification accuracy in terms of the number of generated features for each class on the validation set, which helps us to determine the best numbers of features to generate for different datasets. We can observe that the performance keeps boosting with the increase of the number of generated features at the beginning, and then it turns stable. The logistic regression is used as the baseline method. The baseline performance from 2-shot to 11-shot works as the upper bound of our method with the generated samples from 1 to 10 (along with the support samples, there are 2 to 11 samples per class) at the 1-shot case, while for the 5-shot case, the baseline performance from 6-shot to

Table 2 The few-shot classification accuracy results on CIFAR-FS and CUB datasets

		CIFAR-FS 5-way		CUB 5-way	
Model	Backbone	1-shot	5-shot	1-shot	5-shot
In.					
MatchingNet Vinyals et al. (2016)	ResNet-12	—	—	72.40	83.60
ProtoNet Snell et al. (2017)	ResNet-12	72.20 ± 0.70	83.50 ± 0.50	71.90	87.40
Curvature Generation Gao et al. (2021)	ResNet-12	73.00 ± 0.70	85.80 ± 0.50	74.66 ± 0.21	88.37 ± 0.12
Meta Navigator Zhang et al. (2021)	ResNet-12	74.63 ± 0.91	86.45 ± 0.59	—	—
Centroid alignment Afrasiyabi et al. (2020)	ResNet-18	—	—	74.22 ± 1.09	88.65 ± 0.55
TriNet* Chen et al. (2019)	ResNet-18	—	—	69.61 ± 0.46	84.10 ± 0.35
MultiSem* Schwartz et al. (2022)	Dense-121	—	—	76.10	82.90
MetaQDA Zhang et al. (2021)	WRN-28-10	75.83 ± 0.88	88.79 ± 0.75	—	—
TasNet* Pan et al. (2023)	WRN-28-10	80.94 ± 0.71	87.44 ± 0.58	80.57 ± 0.73	89.89 ± 0.48
DC+LR Yang et al. (2020)	WRN-28-10	—	—	79.56 ± 0.87	90.67 ± 0.35
SIFT+LR* (ours)	WRN-28-10	81.35 ± 0.75	89.22 ± 0.54	82.68 ± 0.75	91.74 ± 0.40
SIFT+SVM* (ours)	WRN-28-10	79.72 ± 0.78	88.31 ± 0.58	84.00 ± 0.71	91.12 ± 0.41
Tran.					
ICI Wang et al. (2020)	ResNet-12	79.19 ± 0.63	86.66 ± 0.36	—	—
ProtoComNet* Zhang et al. (2021)	ResNet-12	—	—	93.20 ± 0.45	94.90 ± 0.31
LaplacianShot Ziko et al. (2020)	ResNet-18	—	—	80.96	88.68
TIM Boudiaf et al. (2020)	ResNet-18	—	—	82.20	90.80
SSR Shen et al. (2021)	WRN-28-10	81.60 ± 0.60	86.0 ± 0.40	—	—
AIM Lee et al. (2021)	WRN-28-10	80.20 ± 0.55	87.34 ± 0.36	—	—
TasNet* Pan et al. (2023)	WRN-28-10	83.68 ± 0.71	89.56 ± 0.58	84.21 ± 0.71	91.62 ± 0.57
SIFT+LR* (ours)	WRN-28-10	85.69 ± 0.82	89.60 ± 0.60	84.94 ± 0.72	91.76 ± 0.42
SIFT+SVM* (ours)	WRN-28-10	84.74 ± 0.79	89.24 ± 0.56	86.00 ± 0.67	91.04 ± 0.42

The best results are highlighted in bold

15-shot is the upper bound of our method with the generated samples from 1 to 10 (along with the support data, there are 6 to 15 samples per class). It can be noticed that, when one extra sample is added, our method achieves almost the $(K+1)$ -shot performance on the baseline classifier, K is the number of labeled samples per class on support set. To further prove that our data augmentation method can indeed work, we design an experiment named “Without SIFT”, which borrows data directly from base classes that are similar to the corresponding novel classes without any transformation. As the number of base samples added to the novel classes increases, the performance degrades drastically (show in blue curve), indicating that adding samples from similar base classes directly will descend the performance of the classifier.

4.4.3 Comparison on Different Classifiers

Figure 4 shows the comparison about the recognition accuracy with respect to different classifiers. Comparing with the distribution calibration method (DC), our SIFT method consistently yields performance gain on different classifiers (i.e. LR, SVM and KNN), which further validates the effective-

ness of our data augmentation technique. We further conduct an experiment on a shallower feature extractor ResNet-12. Results shown in Table 3 demonstrate that our method is applicable for the shallower feature space as well.

4.4.4 Results on Different Number of Augmented Features

Table 4 presents the quantitative results of our SIFT method at different number of generated features. LR denotes the logistic regression method, which is used as the classifier to evaluate our data augmentation technique. The accuracies shown in the first line denote the performances of LR at different support shots, while for the other lines, they present the performance of the data augmentation-based methods at different number of augmented samples. The results shown in Table 4 illustrate that our SIFT method is better than DC (Yang et al., 2020) method in accuracy at different number of augmented samples. “Without SIFT” is a method that we design to prove our semantic transformation process is the reason for the performance boosting. In this experiment, we borrow samples directly from the base classes (which is semantically similar to the corresponding novel

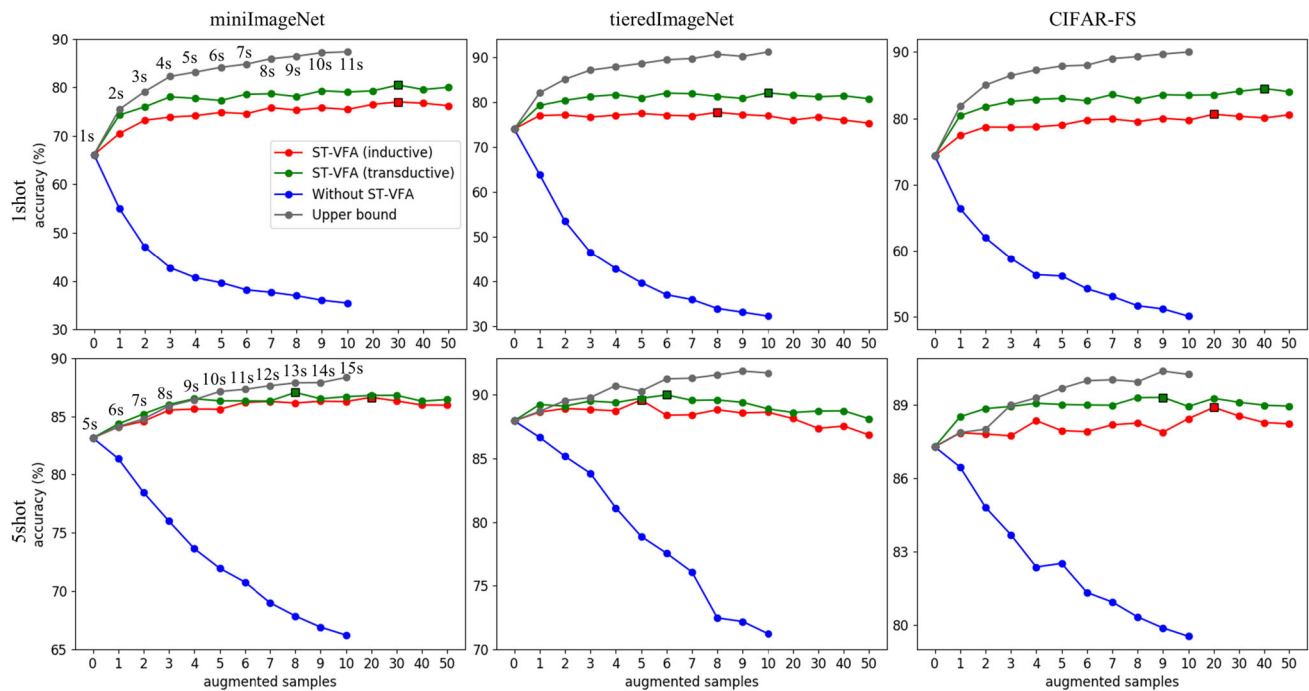


Fig. 3 Illustration of the performance change with varied number of augmented samples on miniImageNet, tieredImageNet and CIFAR-FS datasets. The square markers with black edge denote the best performance points. All the classification is implemented with the logistic regression (LR) classifier

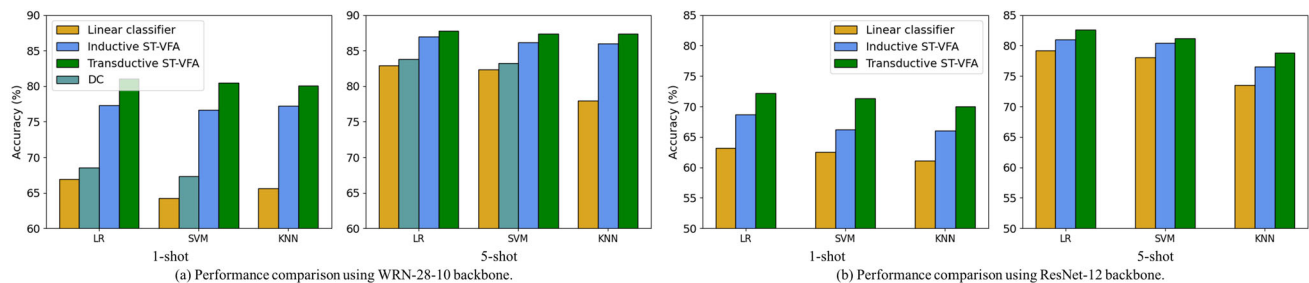


Fig. 4 Comparison of our SIFT method at different classifiers and backbones on miniImageNet dataset

Table 3 The few-shot classification accuracy using ResNet-12 as the feature extractor

	miniImageNet		tieredImageNet		CIFAR-FS	
Model	1-shot	5-shot	1-shot	5-shot	1-shot	5-shot
LR (baseline)	63.17 ± 0.83	79.17 ± 0.58	71.23 ± 0.91	84.84 ± 0.61	72.86 ± 0.90	84.54 ± 0.66
SIFT+LR ^I	68.72 ± 0.73	81.00 ± 0.58	72.15 ± 0.90	85.62 ± 0.65	77.65 ± 0.75	86.42 ± 0.58
SIFT+LR ^T	72.21 ± 0.91	82.56 ± 0.57	77.38 ± 0.92	86.93 ± 0.59	81.93 ± 0.86	87.07 ± 0.60
SVM (baseline)	62.54 ± 0.79	78.02 ± 0.59	70.66 ± 0.90	84.29 ± 0.66	72.80 ± 0.90	84.52 ± 0.64
SIFT+SVM ^I	66.18 ± 0.72	80.40 ± 0.57	71.72 ± 0.95	85.75 ± 0.62	76.17 ± 0.84	86.10 ± 0.62
SIFT+SVM ^T	71.36 ± 0.82	81.16 ± 0.58	76.07 ± 0.95	85.86 ± 0.61	81.33 ± 0.88	86.84 ± 0.61

The superscripts denote the settings: ^I Inductive, ^T Transductive

Table 4 Performance comparison on different number of augmented samples on miniImageNet, where features are extracted with the WRN-28-10 backbone

Shot	Methods	Accuracy (%) on various number of generated samples/shots						
		1	5	10	20	30	40	50
1shot	LR (baseline)	66.73	83.14	86.76	89.10	90.03	90.49	90.97
	Without SIFT	55.01	39.66	35.43	32.82	32.02	31.65	30.68
	DC+LR Yang et al. (2020)	61.31	66.49	66.72	66.21	67.06	67.74	67.31
	SIFT+LR(In)	70.46	74.83	75.43	76.47	77.31	76.71	76.21
	SIFT+LR(Tran)	73.81	78.64	78.98	79.47	81.03	79.81	79.72
5shot	Without SIFT	81.37	71.97	66.24	60.63	56.57	54.94	53.31
	DC+LR Yang et al. (2020)	83.0	82.22	83.24	83.0	83.15	83.49	83.30
	SIFT+LR(In)	84.39	85.60	86.26	86.95	86.31	85.96	85.95
	SIFT+LR(Tran)	84.69	86.69	87.33	86.80	86.82	86.59	86.10

classes) without any transformation. Results of this experiment show drastic performance degradation as the number of base samples increases, which demonstrates that the semantic transformation process is essential, thus proves that our data transformation technique can indeed work.

4.4.5 t-SNE Visualization of the Generated Features

To qualitatively demonstrate the effectiveness of our approach, we show the t-SNE visualization (Van der Maaten & Hinton, 2008) of the generated features in Fig. 5. It can be observed that the generated features exhibit clear clustering structure, and is compatible with the support and query samples, which can help train a discriminative classifier. We also compare the rectified prototypes (denoted as the transductive prototypes) with the support prototypes (denoted as the inductive prototypes) in the last column of Fig. 5. It can be observed that the transductive prototypes exhibit closer to the centers of the query set than the inductive ones, demonstrating the effectiveness of our prototype rectification method in reducing bias.

4.4.6 Selection of Hyperparameters

To select the best hyperparameters for our model, we perform traversal experiments about λ_1 , λ_2 and λ_3 on the validation dataset. We use variable-controlling method to show the performance change under different values. λ_1 (λ_2 and λ_3) is varied from 0.1 to 1 with 0.1 as interval, while the other two parameters are fixed to 1. From Fig. 6 we can observe that the performance in terms of λ_1 , λ_2 and λ_3 keeps a trend of growing as the increase of λ values in 1-shot case. While for the 5-shot case, the performance in terms of the varying values of λ_1 and λ_2 keeps smooth in 5-shot case. The growing trend in terms of the increasing value of λ_3 is especially

obvious for both 1-shot and 5-shot cases. This experiment demonstrates that setting λ_1 , λ_2 and λ_3 to 1.0 is reasonable.

4.5 Ablation Study

4.5.1 Ablation Study on the Constraints

To study the impact of each constraint on the final performance, we ablate the loss terms: the reconstruction loss of the encoder (the second term of Eq. (3), which we denote it as $\mathcal{L}_{enc2}$), the reconstruction loss of the decoder (the second term of Eq. (5), which we denote it as $\mathcal{L}_{dec2}$), the compactness loss $\mathcal{L}_{cpt}$ and all these three constraints, respectively. The projection constraints of the encoder and decoder are the skeletons of our framework, which cannot be ablated. From Table 5, we can observe performance decline when any constraint is ablated. The degradation is especially severe in the 1-shot case.

4.5.2 t-SNE Visualization of the Generated Features Under Ablation Study

We further qualitatively analyze the impact of the constraints on the model by showing the t-SNE visualization of the generated features under ablation study. From Fig. 7, we can observe that the generated features show a deviation to the features extracted from images in the novel task, when any constraint is ablated. As analyzed before, without the reconstruction constraint on the encoder, it will lose some important information contained in the original visual features during the projection from visual to semantic space. While without the reconstruction constraint on the decoder, it will produce semantically meaningless features. Taking a closer look at the t-SNE visualization in third line, we can observe that the features generated without the compactness loss exhibit more biased than that without the encoder or

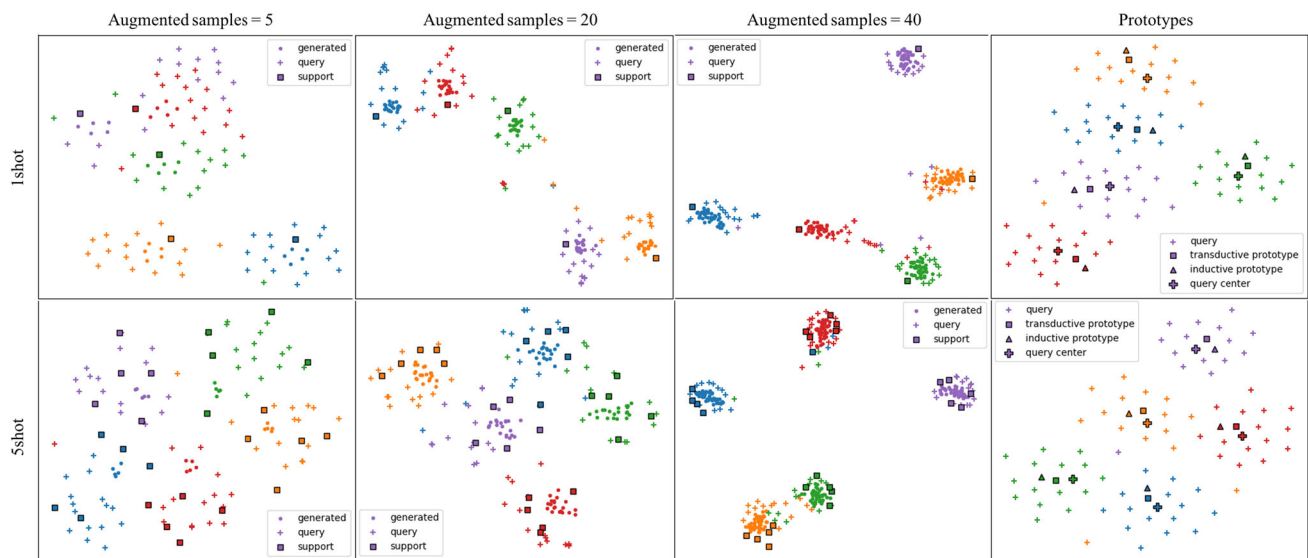


Fig. 5 t-SNE visualization of the augmented samples. The transductive prototype is computed with Eq. (9), while the inductive prototype is computed by averaging samples of each class from the original support set. Features are extracted with WRN-28-10 backbone

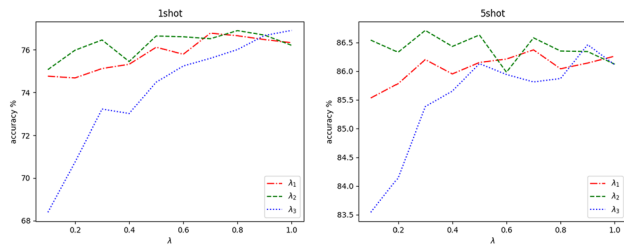


Fig. 6 Performance change about the classification accuracy in terms of the hyperparameters λ_1 , λ_2 and λ_3 . The experiment is conducted on miniImageNet dataset with WRN-28-10 as the feature extractor

decoder reconstruction constraint, which indicates that the compactness constraint contributes more to the final performance. Furthermore, when these three constraints are removed from the optimization objective, the generated features are far away from the support and query samples. The qualitative results are in consistent with the quantitative results in Table 5. The visualizations prove that these constraints can indeed help reduce the distribution bias, resulting in valid augmented samples.

Table 5 Ablation study on the constraints for both inductive and transductive settings in miniImageNet with WRN-28-10 as the backbone

			miniImageNet 5-way			
			In	Tran		
$\mathcal{L}_{enc2}$	$\mathcal{L}_{dec2}$	$\mathcal{L}_{cpt}$	1-shot	5-shot	1-shot	5-shot
✓	✓	✓	77.31 ± 0.67	86.95 ± 0.53	81.03 ± 0.71	87.80 ± 0.47
	✓	✓	71.57 ± 0.76	84.75 ± 0.52	75.98 ± 0.82	85.75 ± 0.53
✓		✓	76.17 ± 0.85	85.92 ± 0.47	78.81 ± 0.79	86.34 ± 0.45
✓	✓		69.25 ± 0.81	84.64 ± 0.50	74.97 ± 0.88	85.99 ± 0.56
			64.45 ± 0.80	82.71 ± 0.56	64.70 ± 0.80	83.10 ± 0.54

The best results are highlighted in bold

4.5.3 Application of our SIFT on Other Baselines

To further test the generalization of our data augmentation technique on other baseline methods, we apply our SIFT on two few-shot classification algorithms: prototypical network (Snell et al., 2017) and MTL (Sun et al., 2019). Results in Table 6 show that the baseline methods equipped with our SIFT technique outperform the baselines by significant margins, and the boosting in the 1-shot case is especially considerable, which demonstrates that the baseline methods can benefit from training with the augmented data generated by our SIFT method.

4.5.4 Performance Comparison of Different Depth Networks

To explicitly reveal the impact of different depth networks on the performance of our SIFT model, we try non-linear networks for the encoder, transformation network and the decoder, respectively, where each of them is composed of two

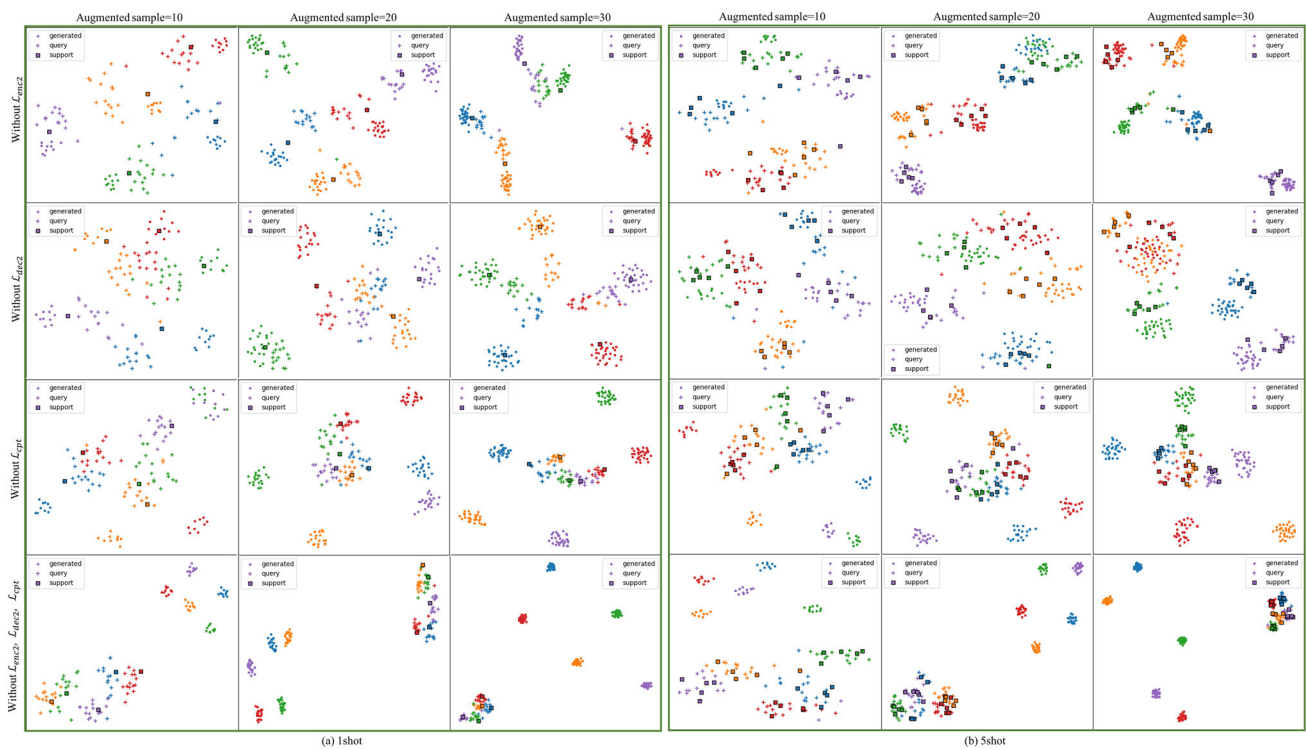


Fig. 7 t-SNE visualization of the ablation study on the constraints: the reconstruction loss of encoder $\mathcal{L}_{enc2}$, the reconstruction loss of decoder $\mathcal{L}_{dec2}$, the compactness loss $\mathcal{L}_{cpt}$, the detailed meanings of these losses

are explained in Sect. 4.5.1. Experiments are conducted on a 5-way 1-shot and 5-shot tasks, where samples are from miniImageNet dataset

Table 6 Performance comparison on ProtoNet (Snell et al., 2017) and MTL (Sun et al., 2019) baselines with and without SIFT technique

Backbone	Methods	Without SIFT 5-way		With SIFT 5-way	
		1-shot	5-shot	1-shot	5-shot
ResNet-12	ProtoNet	61.35 ± 0.87	79.54 ± 0.58	63.99 ± 0.82	80.17 ± 0.54
	MTL	60.82 ± 0.80	75.96 ± 0.62	63.18 ± 0.80	76.92 ± 0.65
WRN-28-10	ProtoNet	65.50 ± 0.81	82.95 ± 0.56	76.45 ± 0.73	86.62 ± 0.49
	MTL	65.57 ± 0.80	79.76 ± 0.60	73.69 ± 0.78	83.09 ± 0.56

The improved results are highlighted in bold

Experiment is conducted on miniImageNet dataset

fully connected layers and one activation function. We compare the performance of the model constructed with linear shallow networks and non-linear deeper networks. As can be seen in Table 7, the non-linear network performs worse than the linear one. Since the training data is very scarce, deeper networks will introduce more parameters, which makes the model easy to overfit the limited training data.

4.5.5 Comparative Performance Analysis with Various Transfer Clues

Our method has achieved remarkable performance through semantic transformation. To investigate the factors contributing to the superior performance of our semantic-based feature transfer strategy, we conduct an ablation study with a focus

Table 7 Performance comparison of the model constructed with linear networks vs non-linear networks in inductive setting on miniImageNet dataset

Model	miniImageNet 5-way	
	1-shot	5-shot
SIFT+LR(linear)	77.31 ± 0.67	86.95 ± 0.53
SIFT+LR(nonlinear)	70.66 ± 0.75	84.32 ± 0.50

Features are extracted with WRN-28-10 backbone

on the transfer clues, aiming to reveal whether visual feature transfer clues can also yield favorable results. For comparison, we employed visual features as the transfer clues to directly transform base features into novel features. In this experiment, we select base classes based on the similarities

Table 8 Comparison of results with different transfer clues

Model	miniImageNet 5-way	
	1-shot	5-shot
LR (baseline)	64.72 ± 0.56	82.36 ± 0.45
DC	68.57 ± 0.55	82.88 ± 0.42
Ours (<i>w/o</i> semantics)	63.59 ± 0.64	82.56 ± 0.50
Ours (<i>w/</i> semantics)	77.31 ± 0.67	86.95 ± 0.53

The experiment is conducted on miniImageNet dataset in inductive setting using WRN-28-10 as the backbone. The table presents the mean accuracy on 600 novel tasks with a 95% confidence interval

between the mean features of the base classes and the prototypes of the novel classes in the visual space. The similarity is computed as: $q_{ij} = \frac{p_i^n p_j^b}{\|p_i^n\| \cdot \|p_j^b\|}$, p_i^n is the prototype of i^{th} class computed by averaging the features in the support set of the novel task, $i = 1, \dots, N$. p_j^b is the mean feature of j^{th} base class, $j = 1, \dots, |C_{base}|$.

Then we select the base classes using the 0–1 LP model as detailed in Eq. (1) and sample features with Eq. (2). All the selected base features are included in X^{bs} . These selected base features are assigned labels corresponding to their respective novel classes. Subsequently, we transform these base features to novel classes directly with a 2-layer fully connected network $h(\cdot|\varphi)$:

$$\hat{x}^n = h(x^b|\varphi), \quad x^b \in X^{bs} \quad (12)$$

where φ is the parameters of the network. $h(\cdot|\varphi)$ is actually the combination of the encoder and decoder used in our method. The input size, hidden size and output size are 640, 300, 640, respectively. $h(\cdot|\varphi)$ is trained with the following loss:

$$\ell = - \sum_{i=1}^{|X^{bs}|} \log \frac{\exp(\hat{x}_i^n \cdot p_{y_i}^n / \tau)}{\sum_{k=1}^N \exp(\hat{x}_i^n \cdot p_k^n / \tau)} \quad (13)$$

where τ is the temperature parameter, which is set to 1 for this experiment. $\hat{x}_i^n$ is the augmented novel feature, with y_i as the label. Then we train a Logistic Regression (LR) classifier using the augmented novel features and the support features of the novel task. The results are reported as the average accuracy over 600 tasks in Table 8.

The results in Table 8 show that the performance of the direct visual feature transfer method, without leveraging semantic clues, deteriorates dramatically to the point where it even underperforms the baseline. This observation underscores the ineffectiveness of direct visual feature transfer without using semantic relationships between classes, thus affirming the pivotal role played by semantic embeddings in ensuring effective feature transfer. Additionally, our explicit

feature transfer strategy demonstrates a notably superior effectiveness compared to the distribution transfer strategy.

4.6 Comparison of Running-time

Previous data augmentation methods have traditionally focused on learning sample distributions and generating a substantial number of samples from these distributions. However, these approaches can be both memory and time-consuming. In contrast to these methods, our SIFT achieves significant performance improvements by augmenting only a few samples, which not only saves time but is also highly sample-effective. To provide insight into the efficiency of our method, we conduct an experiment to measure the per-task running time at various numbers of generated samples and compared our SIFT with the DC method (Yang et al., 2020). This experiment is conducted in miniImageNet dataset. From Table 9, we can observe that in the 1-shot case, the running time of our SIFT method is comparable to that of the DC method when the number of augmented samples is less than 20. However, as the number of augmented samples increases, our SIFT method surpasses the DC method in terms of running time. Conversely, in the 5-shot case, the DC method demands significantly more time compared to our SIFT method. The substantial runtime requirement of the DC method in the 5-shot case is attributed to its approach of generating samples for each support sample individually rather than by class. That is the computational time is directly influenced by the “shot” of the task. For example, in the 5-way 5-shot scenario, the DC method needs to compute 5×5 distributions and generate $25 \times \text{num_aug}$ samples (where num_aug represents the number of augmented samples) for each task. This process substantially increases the computational burden. In contrast, our method augments samples by class, which is agnostic to the “shot” of the task. Overall, the running time increases as the number of augmented samples grows. All the experiments are run on the GTX TITAN X GPU.

5 Conclusion

In this study, we introduce a straightforward yet effective data augmentation method to address the extremely low-shot classification problem. Our SIFT method facilitates the transfer of samples from base classes to novel tasks through semantic transformation to augment features for the novel classes. Since the novel class prototypes serve as anchors for the generated samples, we propose a shortest path optimization technique-based prototype rectification approach to incorporate the cluster centers with the initial prototypes, aiming to mitigate prototype bias. Subsequently, we train a linear classifier using both the support features and the aug-

Table 9 Comparison of per-task running times between our SIFT method and the DC method in miniImageNet dataset, under 5-way 1-shot and 5-way 5-shot tasks

Setting	Methods	Run time (s/task) under different generated samples						
		1	5	10	20	30	40	50
Tran	SIFT+LR(1 s)	3.3	3.5	3.8	4.7	5.3	5.9	6.9
	SIFT+LR(5 s)	3.6	3.9	3.9	4.8	5.6	6.0	6.9
	SIFT+LR(1 s)	3.2	3.6	3.7	4.6	5.4	5.9	6.7
In	SIFT+LR(5 s)	3.7	3.9	4.0	4.9	5.6	6.1	7.2
	DC+LR(1 s)	3.4	3.3	3.6	3.7	3.8	3.7	3.8
	DC+LR(5 s)	9.5	9.2	9.4	9.6	9.6	10.1	11.8

“Tran” and “In” indicate transductive and inductive settings, respectively

“1 s” and “5 s” denote 1-shot and 5-shot, respectively

mented features generated through our SIFT method, which is then employed to predict labels for the query samples. The extensive experiments and comprehensive ablation studies consistently demonstrate the superior performance and robust generalization of our proposed SIFT method.

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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